

Introduction to Computer Network

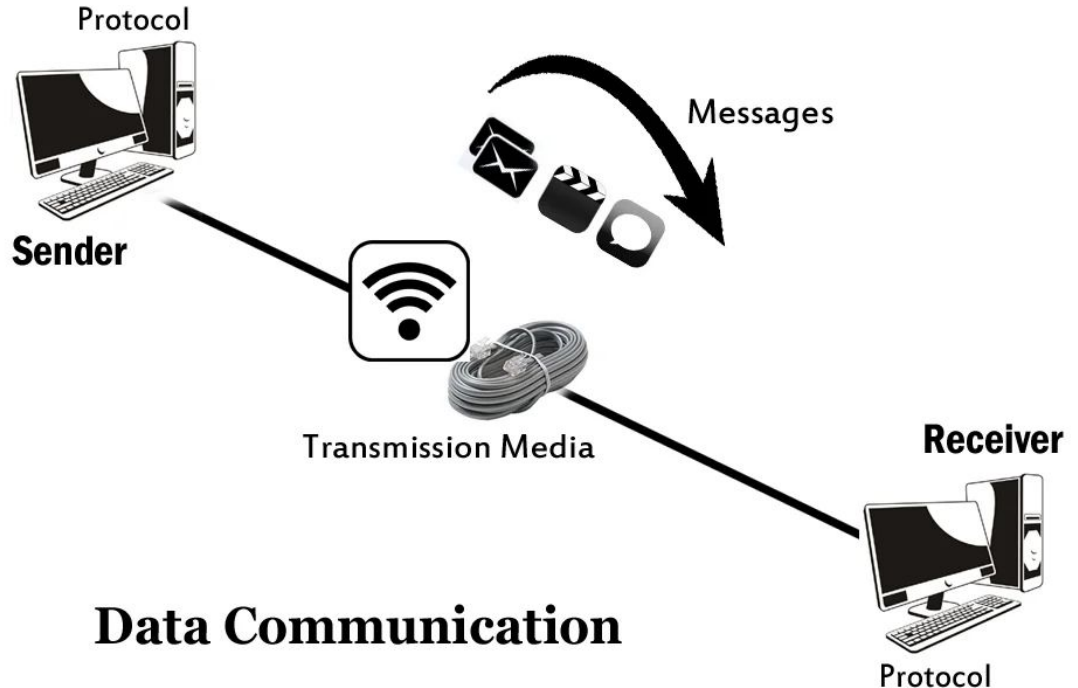
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What is Computer Network?

A Computer Network is a collection of **interconnected** devices that exchange information and share resources using a common **set of rules** over a **communication medium**.

Data Communication

- **Sender**
- **Receiver**
- **Message**
- **Medium**
- **Protocol**



Effectiveness of Data Communication

1. **Delivery** – The system must deliver data to the correct destination, ensuring it reaches only the intended device or user.
2. **Accuracy** – The system must deliver data without errors, so that the received information matches exactly what was sent.
3. **Timeliness** – The system must deliver data within an acceptable time, so it remains useful for the intended application.
4. **Jitter** – The system must maintain consistent packet delivery timing, avoiding uneven delays that can affect real-time applications like audio or video.

Data Communication/Transmission Modes

1. **Simplex** – Data flows only in one direction; the sender can transmit, and the receiver can only receive.
Example: Keyboard to CPU, Television broadcast.
2. **Half-Duplex** – Data flows both ways, but not at the same time; the sender and receiver take turns transmitting.
Example: Walkie-talkie,
3. **Full-Duplex** – Data flows simultaneously in both directions; sender and receiver can communicate at the same time.
Example: Telephone conversation, Chats

Core Components of Network Infrastructure

- **End Device**
- **Transmission Media**
- **Protocol**
- **Networking Device**

End Devices

- Devices where communication starts or ends.
- Generate digital data that needs to travel over a network.
- Each device has a unique address for identification.
- Run applications that use network services

Example: Phones, Laptops

Transmission Media

- Provides the physical or wireless path that carries data.
- Can be wired (copper, fiber) or wireless (Wi-Fi, mobile signals).
- Determines the speed, distance, and reliability of data transfer.
- Supports multiple devices transmitting data simultaneously.

Example: Ethernet Cable, Wifi

Networking Devices

- Connect multiple end devices and networks together.
- Forward data toward its intended destination using addresses.
- Manage traffic to avoid congestion and ensure smooth delivery.
- Enable communication between local networks and distant networks.

Example: Router, Switch

Protocols

- Define the rules and standards for communication.
- Specify data formats, addressing, and error detection.
- Ensure reliable and understandable communication between devices.
- Allow devices from different manufacturers to interoperate seamlessly.

Example: HTTP, FTP

Application of Computer Network

- Communication
- Resource Sharing
- Data Access & Storage
- E-Commerce & Online Service
- Collaboration
- Entertainment & Multimedia
- Remote access and Teleworking

Network Architecture

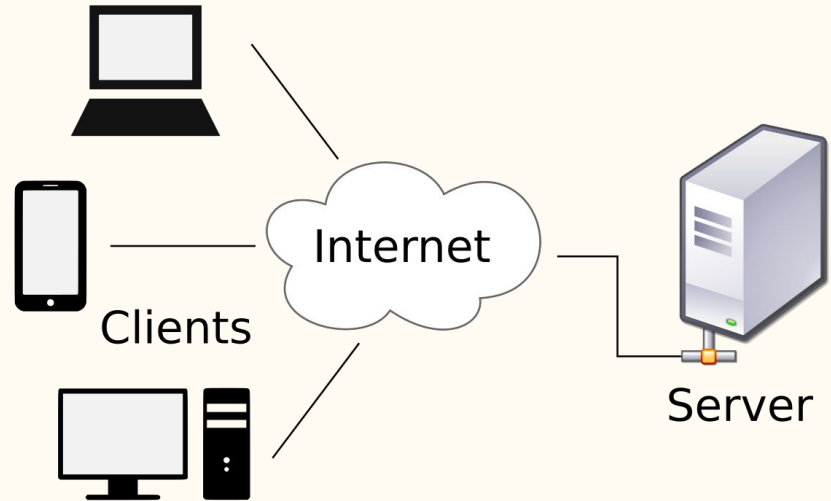
Network architecture is the structured design of a computer network, specifying its components, their roles, connections, and communication rules to ensure reliable and efficient data transfer.

Types:

- ***Client Server***
- ***Peer to Peer***

Network Architecture: Client Server

A network architecture where a central server provides resources or services to multiple client devices that request them.



Network Architecture: Client Server

- **Single source of truth**

All important data lives in one place, so clients always see the same and up-to-date information.

- **Central decision-making**

The server decides who can access what, when, and how.

- **Simpler clients**

Clients mainly send requests and display results. All heavy processing happens on the server.

- **Controlled environment**

Security rules, backups, and updates are applied once on the server instead of on every device.

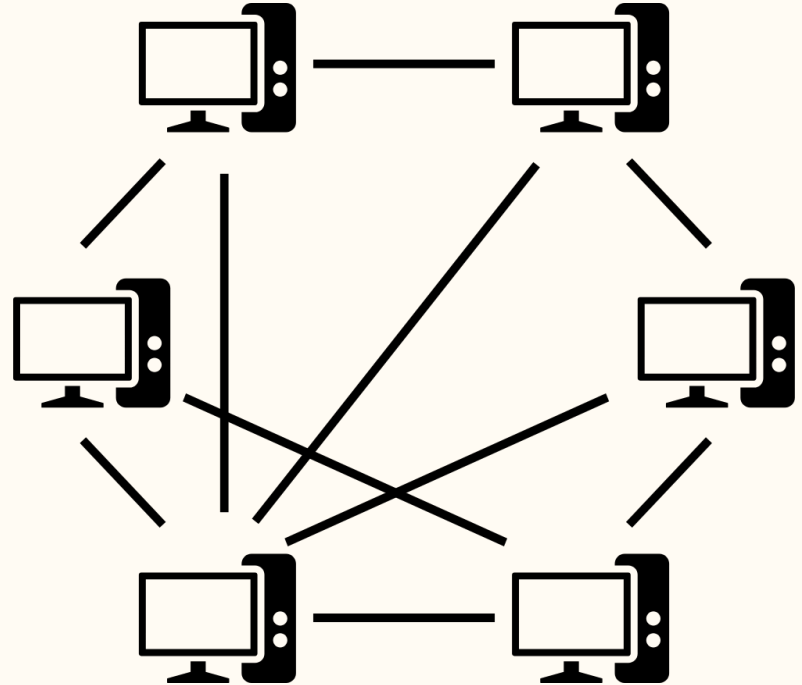
Network Architecture: Client Server

Disadvantages:

1. Single point of failure.
2. Expensive to set up and maintain (server hardware/software).
3. Can create network bottlenecks if server is overloaded.

Network Architecture: Peer 2 Peer

A peer-to-peer network is a network architecture where all devices (peers) are equal, and each can act as both client and server, sharing resources directly.



Network Architecture: Peer 2 Peer

- **No boss node**

There is no single system controlling the network and every peer has equal authority.

- **Direct sharing**

Devices exchange data directly instead of going through a central server.

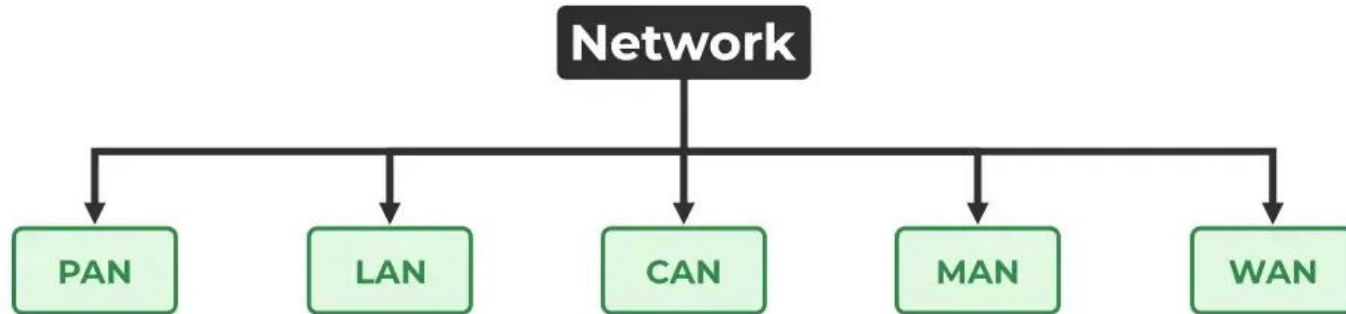
- **Workload is spread out**

Storage, bandwidth, and processing are shared among all peers.

- **Failure is less disruptive**

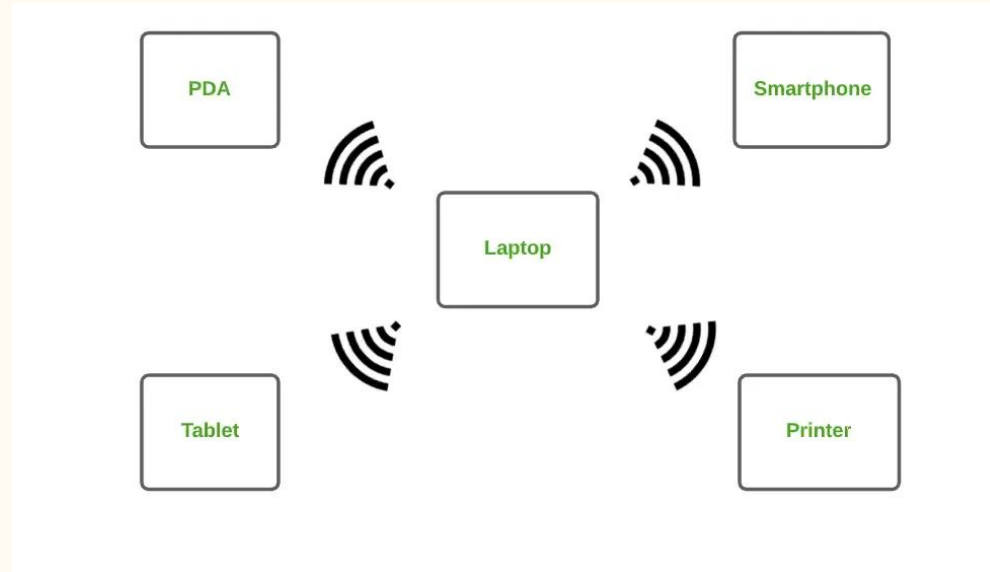
Losing one peer does not shut down the entire network.

Types of Computer Network



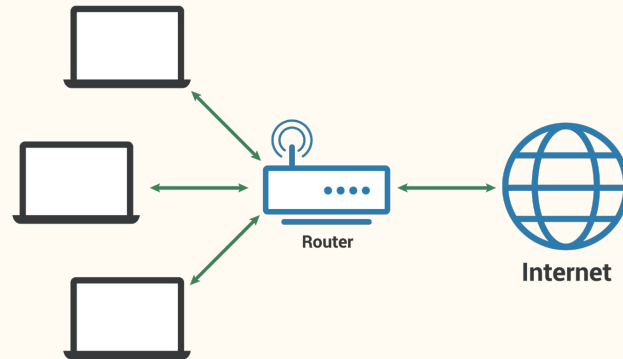
Personal Area Network (PAN)

A Personal Area Network (PAN) is a short-range network designed for communication among devices located around an individual user. It ranges from few centimeters to ~10 meters



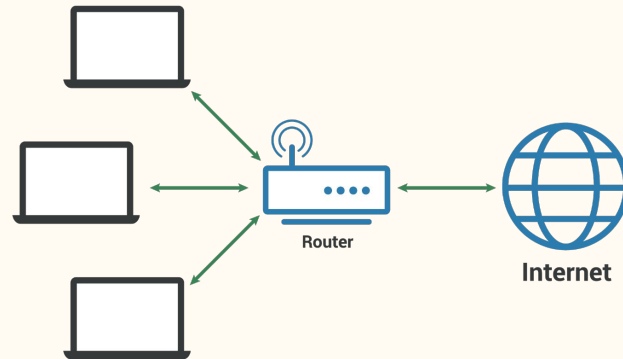
Local Area Network (LAN)

A Local Area Network (LAN) interconnects computers within a limited geographic area such as a building, home, or campus.



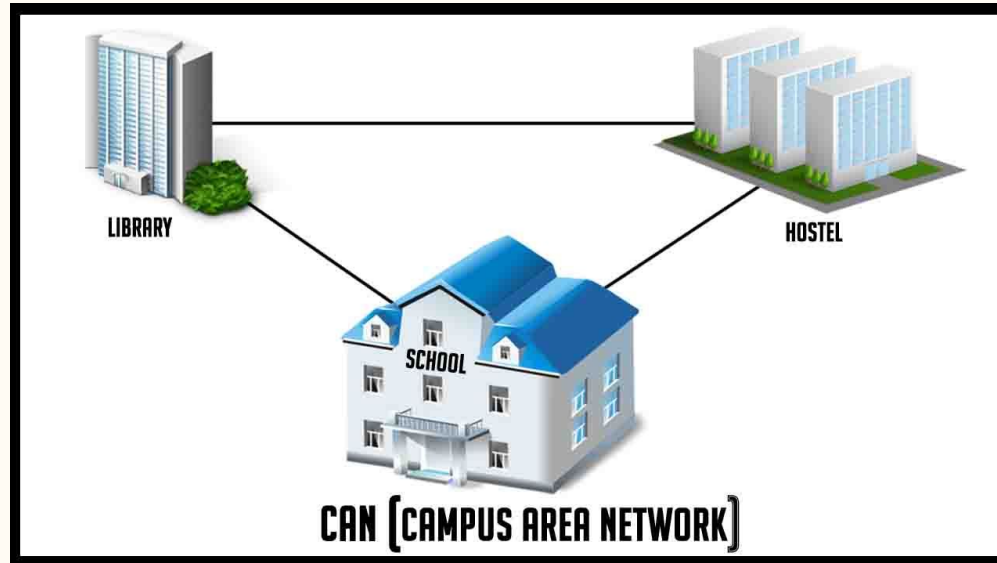
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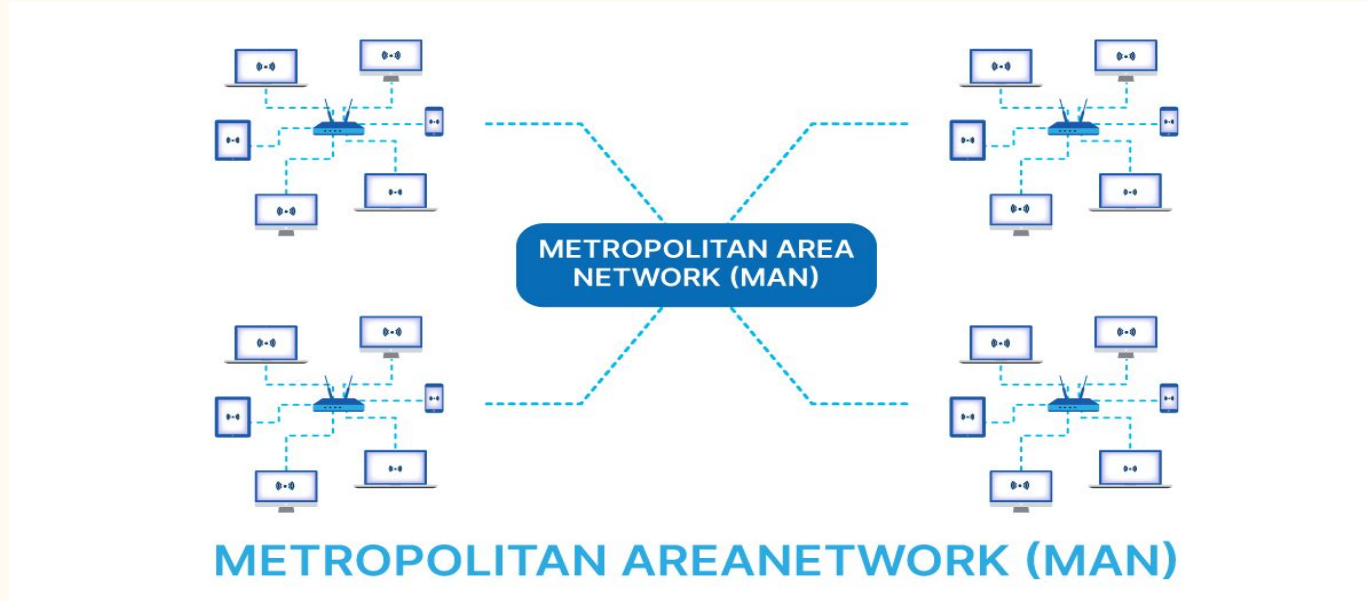
Campus Area Network (CAN)

A Campus Area Network (CAN) is a network that interconnects multiple LANs within a limited geographical area such as a university campus or corporate park



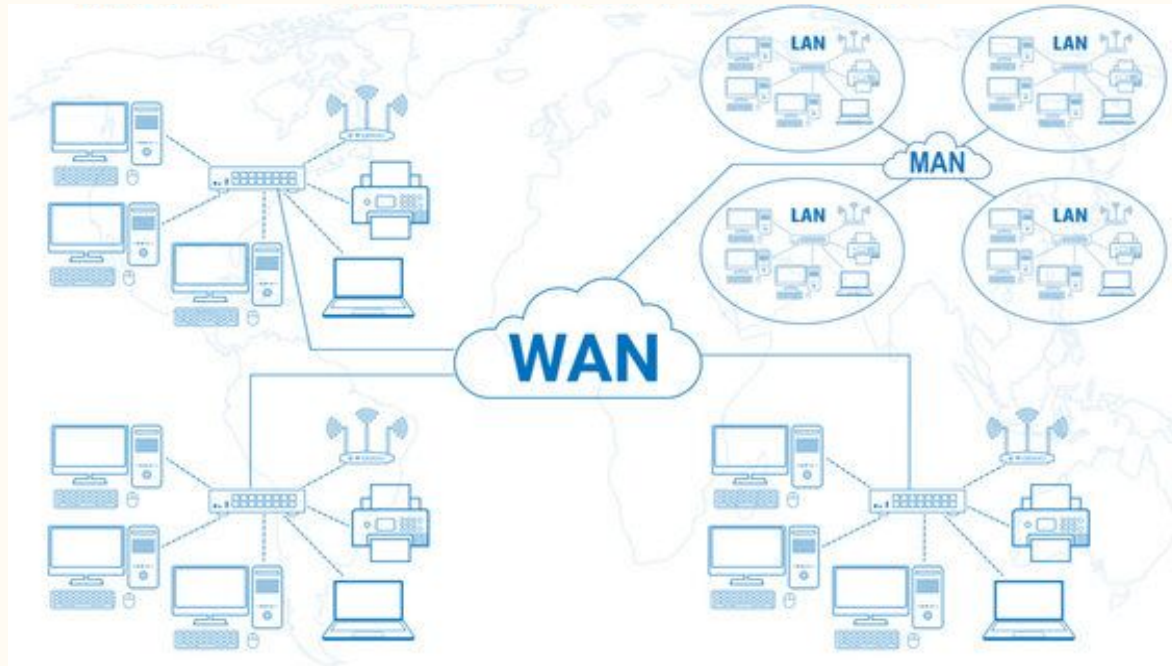
Metropolitan Area Network (MAN)

A Metropolitan Area Network (MAN) connects multiple LANs across a city or metropolitan region.



Wide Area Network (WAN)

A Wide Area Network (WAN) spans large geographic distances of countries or continents and connects multiple LANs and MANs.



Protocol And Standard

A **protocol** is a set of rules that defines how two or more devices communicate over a network.

A protocol answers three fundamental questions:

1. *When does communication start?*
2. *What does each bit mean?*
3. *When is communication finished?*

Example: HTTP, FTP, ICMP, DHCP etc.

Protocol And Standard

A **standard** is an agreed-upon specification that ensures different systems and vendors implement protocols in a consistent and compatible way.

If protocols are rules of communication, standards are the official agreement that everyone must follow the same rulebook.

Without standards:

- Vendor A's HTTP would not match Vendor B's HTTP
- Routers from different companies would refuse to cooperate
- The Internet would fracture into incompatible mini-networks

Protocol And Standard

Major Organizations Defining Network Standards:

1. **IEEE (Institute of Electrical and Electronics Engineers)**

Defines standards for local area networks (LAN) and wireless networks (WLAN)—basically how devices physically connect and exchange frames over cables and radio signals.

Example:

- IEEE 802.3 → Ethernet (wired LAN communication)
- IEEE 802.11 → Wi-Fi (e.g., 802.11n, 802.11ac, 802.11ax)

Protocol And Standard

2. IETF (Internet Engineering Task Force)

Defines standards for core Internet communication protocols how data is addressed, routed, transported, and exchanged across the global Internet.

Example:

- RFC 793 → TCP (reliable data transmission)
- RFC 1034/1035 → DNS (domain name resolution)
- RFC 9110 → HTTP semantics (web communication rules)

Protocol And Standard

3. ISO (International Organization for Standardization)

Defines international standards ensuring interoperability and compatibility between networked systems and communication technologies.

Example:

- ISO/IEC 8802 series → LAN/MAN standards aligned with IEEE 802
- ISO/IEC 7498-1 → OSI (Open Systems Interconnection) Reference Model

Open Systems Interconnection (OSI) Model

The OSI model is a framework created by ISO to standardize how devices communicate over a network.

It splits communication into seven layers, each with a clear role.

Each layer of the OSI model interacts with the layer directly above and below it, encapsulating and transmitting data in a structured manner.

Open Systems Interconnection (OSI) Model

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

OSI: Physical Layer

The Physical layer is the first layer and deals with the actual transmission of raw bits over a physical medium, such as cables, fiber optics, or wireless signals.

It defines the hardware aspects of communication, including voltage levels, timing, and connectors.

Devices like network interface cards (NICs), Ethernet cables, and Wi-Fi radios operate at this layer. Without the physical layer, no data can move between devices.

OSI: Data Link Layer

The Data Link layer ensures reliable communication between two directly connected devices.

It organizes raw bits from the physical layer into frames and adds MAC addresses to identify devices on the same network.

This layer also detects and sometimes corrects errors in transmission.

Switches and bridges operate at the data link layer, and Ethernet or Wi-Fi protocols (IEEE 802.3 and IEEE 802.11) are examples of standards implemented at this layer.

OSI: Network Layer

The Network layer is responsible for moving data from the source device to the destination device across multiple networks.

It provides logical addressing using IP addresses and determines the best path for data to travel through routers.

Protocols like IPv4, IPv6, and ICMP function at this layer.

The network layer ensures that data can reach devices located on different networks efficiently.

OSI: Transport Layer

The Transport layer guarantees that data is delivered completely and correctly from one application to another.

It segments large messages into smaller chunks and manages error checking, flow control, and retransmission.

This layer can operate in a connection-oriented mode, like TCP, which ensures reliability, or a connectionless mode, like UDP, which is faster but does not guarantee delivery.

OSI: Session Layer

The Session layer manages the sessions between applications on different devices.

It is responsible for establishing, maintaining, and terminating communication sessions.

Examples include remote procedure calls (RPC) and session management for video calls.

The session layer keeps communication organized between applications.

OSI: Presentation Layer

The Presentation layer translates and formats data so that it can be understood by the application layer.

It handles encryption and decryption to secure communication, compression to reduce data size, and conversion between different data formats.

Protocols and formats like SSL/TLS encryption operate at this layer, ensuring that the sender and receiver understand the data correctly.

OSI: Application Layer

The application layer is the layer closest to the user and makes network communication practical and accessible.

It is the interface through which users interact with the network.

This layer defines protocols for services such as web browsing, email, and file transfer.

Examples include HTTP for web pages, FTP for file transfer, SMTP for email, and DNS for domain name resolution.

TCP/IP

The TCP/IP model (Transmission Control Protocol/Internet Protocol model) is a set of communication protocols used to connect devices over the Internet and other networks.

It defines how data is transmitted, addressed, routed, and received between devices in a real-world network environment.

Unlike the OSI model, which is purely conceptual, the TCP/IP model is based on protocols that are actually implemented and deployed in operating systems, routers, and servers.

TCP/IP

OSI Model vs TCP/IP Model



Critique of OSI Model

1. **Bad Timing:** OSI arrived when TCP/IP was already widely used in universities and research networks. Vendors and organizations were reluctant to adopt a second protocol stack, so OSI remained mostly theoretical.
 2. **Bad Technology:** The model repeated functions like error control and addressing across layers, making it complex and inefficient.
 3. **Bad Implementations:** Early OSI software was large, slow, and resource-heavy, which gave it a reputation for poor performance.
- Bad Politics:** OSI was driven by international committees, while TCP/IP emerged from academic research. Developers preferred TCP/IP, perceiving OSI as bureaucratic.

Critique of TCP/IP

1. **Mixed Concepts:** TCP/IP doesn't separate services, interfaces, and protocols clearly. HTTP in the Application layer both defines the service and the protocol itself.
2. **Not General-Purpose:** TCP/IP works well for Internet protocols but not for others like Bluetooth.
3. **Weak Link Layer:** The Link layer combines physical transmission and frame delivery, making it hard to study hardware-level functions. Ethernet frames (Data Link) and cable signals (Physical) are merged.
4. **No Physical/Data Link Separation:** OSI separates them for clarity. TCP/IP merges them, reducing precision.
5. **Ad Hoc Protocols:** Some TCP/IP protocols were developed quickly by researchers and became entrenched, even if not fully efficient.